τ-Voice

Benchmarking Full-Duplex Voice Agents on Real-World Domains

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ICML 2026

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Can Voice Agents Actually Do the Job?

Full-duplex voice agents listen and speak simultaneously—the next frontier in conversational AI.

But today's benchmarks evaluate capabilities **in isolation**:

- τ^2-bench: complex grounded tasks, but **text-only**
- Full-Duplex-Bench: turn-taking dynamics, but **no real tools**

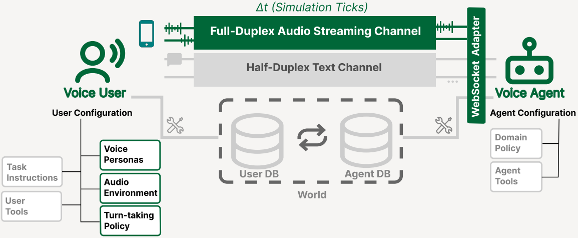
Voice compounds task difficulty:

- No punctuation, fillers, disfluencies
- Accents, background noise, telephony compression
- Interruptions, backchannels, real-time turn-taking

Missing: end-to-end evaluation of voice agents on **grounded tasks with real-world complexity**.

τ-Voice: a Voice Agent Benchmark

Extending τ^2-bench to full-duplex voice interaction



Domains	Retail, Airline, Telecom
Tasks	278 total

Models	`gemini-live-2.5`
	`gpt-realtime-1.5`
	`grok-voice`

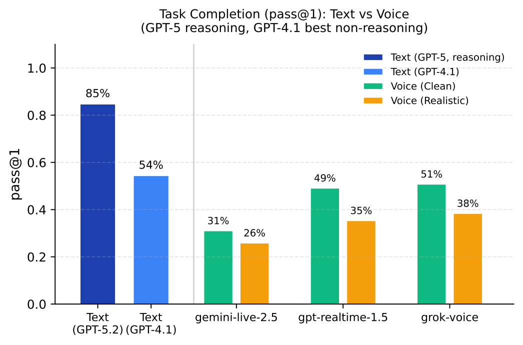
Two conditions:

- **Clean**: clear audio, American accent, no interruptions
- **Realistic**: diverse accents, noise, turn-taking dynamics

+ single-factor ablations on Retail

Headline: a Large Voice–Text Gap Remains

pass@1 averaged across all domains

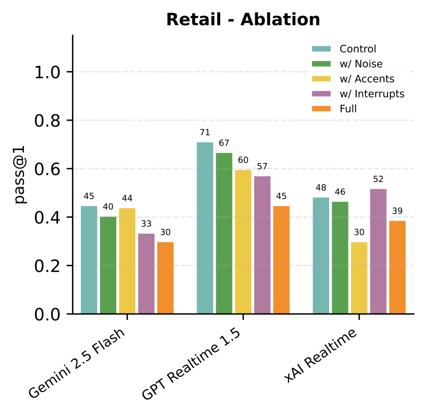


Voice agents retain only **30–45%** of text SOTA capability under realistic conditions.

Key findings:

- Best voice (`grok-voice`, 51%) nearly matches GPT-4.1 (54%) under Clean—but **still 34pp below GPT-5**
- Realistic conditions drop performance a further 12pp on average
- `gemini-live-2.5` most **robust** to degradation (loses only 17% of Clean)
- `grok-voice` slightly **highest absolute** scores
- All gaps are **statistically significant** ($p \leq 0.002$, paired permutation tests)

What Hurts Most? Ablations & Voice Quality



Accents most damaging on avg (−10pp) but highly model-specific.

Voice interaction quality (Realistic)

	Latency↓	Resp.↑	Intrpt.↓	Select.↑
gemini	1.14s	69%	**21%**	54%
gpt-rt	**0.90s**	**100%**	14%	6%
grok	1.15s	83%	84%	**57%**

Each model excels on different dimensions but fails on at least one. **No model masters both task completion and conversational dynamics.**

Takeaways

1. **The voice–text gap is real and large.** Best voice retains only 45% of text SOTA—even under clean conditions.
2. **It's not just ASR.** Logical errors and hallucinations dominate even with clean audio.
3. **Acoustic realism matters unevenly.** Accents are devastating for some models but not others—accessibility concerns.
4. **No model does it all.** Fundamental trade-off between responsiveness and restraint.

Try it:

github.com/sierra-research/tau2-bench

taubench.com — live leaderboard

taubench.com/#trajectory-visualizer

A reproducible testbed for natural, conversational, reliable voice agents.

Thank you.

Questions?

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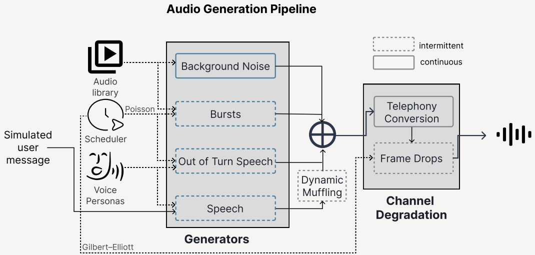
Extension Material

Methods, ablations, error analysis, limitations

Voice User Simulator

Generating realistic caller audio with controllable complexity



- 7 voice personas with diverse accents
- Realistic audio: noise, burst sounds, frame drops, muffling, telephony compression
- LLM-driven turn-taking: interruptions, backchannels, yielding
- Decoupled from wall-clock time → use the most capable LLM without real-time constraints

Domains & Tasks

278 tasks adapted from τ^2-bench, with voice-specific user instructions

- **Retail** (114 tasks) — returns, exchanges, cancellations, order modifications. Heavy slot-filling: names, emails, order IDs.
- **Airline** (50 tasks) — flight changes, seat upgrades, booking modifications with passenger verification.
- **Telecom** (114 tasks) — plan changes, billing, service activations, account modifications.

Retail is the primary evaluation domain—slot filling (names, emails, order IDs) is where end-to-end speech systems are known to struggle.

Apples-to-apples with text: same tasks, same evaluation harness, only the user-side instructions are voice-adapted.

Models in scope: audio-native, full-duplex, native tool calling.

Conditions: Clean, Realistic, and Single-Factor Ablations

What each speech-complexity setting enables

Category	Setting	Clean	+Noise	+Accents	+Turn-taking	Realistic
Accents	Personas			✓		✓
Audio/Channel	Background noise		✓			✓
	Burst noise		✓			✓
	Frame drops		✓			✓
	Telephony	✓	✓	✓	✓	✓
	Muffling		✓			✓
Turn-Taking	Involuntary sounds				✓	✓
	Non agent-directed speech				✓	✓
	Interruptions				✓	✓
	Backchanneling				✓	✓

Realistic combines all factors. Single-factor ablations isolate each contribution on Retail.

Single-Factor Ablation Results (Retail)

pass@1 with each factor added on top of Clean

Condition	`gemini-live-2.5`	`gpt-realtime-1.5`	`grok-voice`	All
Clean	45%	**71%**	48%	55%
+ Noise	40% (–4)	**67% (–4)**	46% (–2)	51% (–4)
+ Accents	44% (–1)	**60% (–11)**	30% (–18)	44% (–10)
+ Turn-taking	33% (–11)	**57% (–14)**	52% (+4)	47% (–7)
Realistic	30% (–15)	**45% (–26)**	39% (–10)	38% (–17)

- **Accents** largest avg drop (–10pp), but **wildly model-specific**: `grok-voice` loses 38% relative; `gemini-live-2.5` barely moves.
- **Turn-taking** is `gemini`'s worst factor (–11pp, –25% rel.).
- Interactions are non-additive and model-specific (e.g. `grok-voice`'s Realistic drop is *smaller* than the sum of individual factors).
- Higher Clean baseline does *not* protect against degradation.

Voice Interaction Quality

Conversational dynamics under Realistic conditions (aggregated across domains)

Model	Latency↓	Responsiveness↑	Interrupt↓	Selectivity↑
`gemini-live-2.5`	1.14s	69%	**21%**	54%
`gpt-realtime-1.5`	**0.90s**	**100%**	14%	6%
`grok-voice`	1.15s	83%	84%	**57%**

- **`gpt-realtime-1.5`**: fastest, most responsive, but *responds to almost everything* (selectivity 6%).
- **`grok-voice`**: best selectivity, but interrupts users nearly once per turn.
- **`gemini-live-2.5`**: lowest interrupt rate, but misses ~1/3 of user turns.

No model achieves both reliable responsiveness and appropriate restraint.

Statistical Significance

Paired permutation tests (100k perms, paired by task ID, Holm-Bonferroni)

Comparison	Model	Rate A	Rate B	Δ (pp)	p (adj)
Text (R) → Clean	`gemini-live-2.5`	85.2%	30.6%	−54.6	<0.001
	`gpt-realtime-1.5`	85.2%	47.8%	−37.3	<0.001
	`grok-voice`	85.2%	52.0%	−33.2	<0.001
Text (R) → Realistic	`gemini-live-2.5`	85.2%	24.3%	−60.9	<0.001
	`gpt-realtime-1.5`	85.2%	33.6%	−51.5	<0.001
	`grok-voice`	85.2%	36.0%	−49.2	<0.001
Clean → Realistic	`gemini-live-2.5`	30.6%	24.3%	−6.3	0.002
	`gpt-realtime-1.5`	47.8%	33.6%	−14.2	<0.001
	`grok-voice`	52.0%	36.0%	−16.0	<0.001

Pooled across all three domains, 2 independent runs per condition. **All headline gaps are significant** at $p \leq 0.002$.

Qualitative Error Analysis

Are failures really the agent's fault—or simulator artifacts?

Cohorts—two failure-source slices:

- **Voice-Fragile** — text passes, Clean voice fails.
- **Noise-Fragile** — Clean passes, Realistic fails.

2 raters annotated 91 failed simulations; 84% initial agreement, 100% after discussion.

Agent errors dominate (79% & 90%): failures reflect agent limitations, not simulator artifacts.

Src	Error Type	V-Frag.	N-Frag.
Agent	Logical	13	16
	Transcription	10	16
	Hallucination	6	6
	VAD/Unresp.	1	4
	Timeout	4	1
	Total	*34 (79%)*	*43 (90%)*
User	Logical	9	1
	Early Term.	0	4
	Total	*9 (21%)*	*5 (10%)*

Failure Mode Details

Common qualitative patterns

Authentication bottleneck:

- Fail to transcribe names/emails even when spelled letter-by-letter
- Blocks downstream actions; compounds with accents

Hallucinated completions:

- "I've updated your address"—no tool call made

Multi-step tracking failures:

- Completes part of a request, forgets the rest

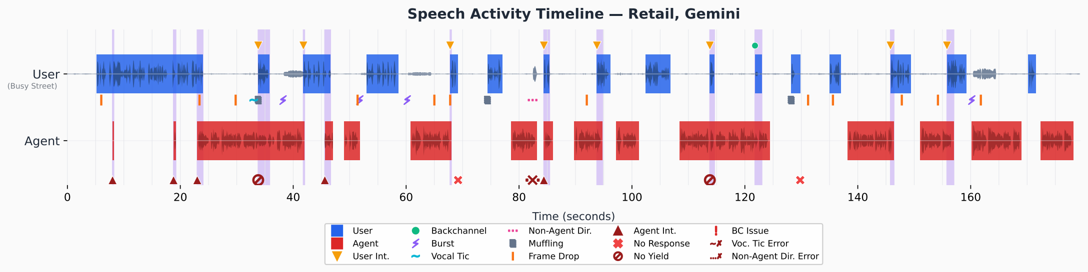
Compounding under realistic conditions:

- Unresponsive after repeated auth failures
- Verbosity causes timeouts on complex tasks

$\sim$95% of agent-attributable failures reflect domain-agnostic conversational skills (spelling, grounding, honesty, tracking, arithmetic)—not domain policy gaps.

Example: Speech Activity Timeline (Task 41, Retail)

Full-duplex interaction with `gemini-live-2.5`—interruptions, backchannels, noise, and agent errors



Blue = user, Red = agent, Purple = overlap. Markers: observations (interruptions, backchannels, noise) and agent errors (no-response, incorrect yield). Outcome: **failed**.

Example: Tick Transcript (Task 41)

Key moments showing simultaneous speech. Orange = agent error. Green = correct. [Brackets] = non-agent-directed.

Time	User	Agent	Event
5–8s 8s	Hi, I have two prob- -lems. First, I ordered...	 Hello!	 *agent interrupts user*
67–68s 68s 69–74s	 Jigsaw first.	...Which would you like to do first?	 *user int., agent yields* *no response for 5 seconds*
82s 84s	[Give me a moment.]	...exchange the puzzle– Sure, take your time.	*non-dir. speech* *error: responds to non-dir.*
113s 114–115s	Yeah, that's it.	...Is that the one? We can exch- -ange it for a puzzle...	*user interrupts* *agent does not yield*
121–122s	mm-hmm	...Would you like to exchange...	*backchannel, correctly continues*

Limitations

Language & speech:

- English-only, TTS-synthesized voices
- Accent findings via TTS personas are *indicative*, not definitive

Evaluation scope:

- Task completion + conversational dynamics
- Not measured: agent speech naturalness, user satisfaction, partial task success

Simulator fidelity:

- More patient than real users, perfect memory, instant tool calls
- Decoupled from wall-clock $\rightarrow$ p95 sim latency $\sim$1.5s
- Human raters scored sim **3.1/4** on naturalness (60 sims; 83% $\geq$3)

Transcript injection:

- ASR bypassed on agent side $\rightarrow$ default mode is an **upper bound** on real-world performance
- Optional ASR-enabled mode confirms the gaps persist

Future Work

Extending the benchmark:

- Tool-call latency metrics
- Agent speech-generation quality
- Non-English languages
- Human user studies to validate simulator dynamics
- Cascaded ASR→LLM→TTS baselines (architecture already supports it)

Living leaderboard:

- Onboarding open-source full-duplex models as they gain native tool calling
- New entrants tracked on taubench.com

Accessibility-focused evaluation:

- Model-specific accent vulnerabilities motivate measuring whether voice agents serve users equitably across accents, speech patterns, and acoustic environments

Backup

Full Results: Text vs Voice (pass@1)

			Voice	
Domain	**Model**	**Text (GPT-5 / GPT-4.1)**	**Clean**	**Realistic**
All	`gemini-live-2.5`	85% / 54%	31% (–54pp)	26% (–59pp)
	`gpt-realtime-1.5`		49% (–36pp)	35% (–49pp)
	`grok-voice`		**51%** (–34pp)	**38%** (–46pp)
Retail	`gemini-live-2.5`	81% / 76%	45% (–36pp)	30% (–51pp)
	`gpt-realtime-1.5`		**71%** (–10pp)	**45%** (–36pp)
	`grok-voice`		48% (–33pp)	39% (–42pp)
Airline	`gemini-live-2.5`	83% / 53%	28% (–55pp)	30% (–53pp)
	`gpt-realtime-1.5`		**48%** (–35pp)	**40%** (–43pp)
	`grok-voice`		46% (–37pp)	36% (–47pp)
Telecom	`gemini-live-2.5`	90% / 34%	20% (–70pp)	18% (–72pp)
	`gpt-realtime-1.5`		28% (–62pp)	21% (–69pp)
	`grok-voice`		**58%** (–32pp)	**40%** (–50pp)

Deltas relative to GPT-5 (reasoning). `grok-voice` leads overall; `gpt-realtime-1.5` leads Retail (71% Clean); domain patterns vary substantially.